

Master’s student in Data Science and Big Data with hands-on experience in machine learning, geospatial analytics, and generative AI. Focused on transforming complex datasets into actionable insights and building intelligent systems that bridge data, AI, and real-world impact.

SKILLS

Programming & Frameworks	Python (pandas, NumPy, scikit-learn, Django), Java (Spring Boot, JEE), JavaScript (React.js, Node.js), HTML, CSS
Machine Learning & AI	Supervised & unsupervised learning, model evaluation, HuggingFace Transformers, NLP, generative AI, API integration
Data Engineering & Big Data	Apache Spark, Hadoop ecosystem, Kafka, Docker, Kubernetes
Databases	SQL (MySQL, PostgreSQL), NoSQL (MongoDB)
DevOps & Tools	Git, CI/CD workflows, Linux (Ubuntu), containerized microservices, RESTful APIs

EXPERIENCE

Data Science Intern <i>WafR</i>	Jul 2025 — Sep 2025 <i>Casablanca</i>
- Designed and implemented a data-driven zoning system to identify and cluster potential grocery store locations across Morocco. - Developed a capacity-constrained clustering algorithm integrating demographic and geographic data to balance zones by store count and population. - Collected, cleaned, and analyzed large-scale geospatial datasets using Python, pandas, geopandas, and scikit-learn. - Created predictive models and interactive Folium maps to support strategic retail expansion decisions.	

PROJECTS

Text-to-Video Generative AI Pipeline <i>Academic Project</i>	Nov 2025 — Present
- Developed a generative pipeline that creates short AI videos from model-generated text using TTS and video synthesis. - Combined text generation (HuggingFace LLMs), text-to-speech models, and automated video assembly tools. - Exploring multimodal AI workflows for storytelling and content generation.	
AI-Powered Educational Dataset & RAG System <i>End-of-Module Project</i>	Mar 2025 — Jun 2025
- Constructed a dataset of Moroccan math exercises via automated web scraping and PDF parsing. - Enhanced data with AI-generated corrections using the DeepSeek API and multilingual embeddings. - Built a semantic search and RAG pipeline with FAISS, HuggingFace models, FastAPI backend, and React frontend.	
Comparative Study of GAN Optimization Algorithms <i>End-of-Module Project</i>	Apr 2025 — Jun 2025
- Conducted empirical analysis of optimization algorithms (Adam, RMSprop, SGD, Lookahead) for GAN training stability. - Implemented DCGAN architecture on CIFAR-10 dataset using PyTorch, evaluating convergence behavior and mode collapse. - Identified Lookahead optimizer as most stable, providing practical recommendations for generative model training.	
Blockchain Certificate Registry <i>End-of-Module Project</i>	Apr 2025 — Jun 2025
- Built a decentralized application (dApp) for registering and managing digital certificates on Ethereum. - Implemented Solidity smart contracts, React frontend, and Truffle + Ganache for local blockchain testing. - Enabled file hashing, ownership transfer, and on-chain verification of digital assets.	
Intelligent Recruitment SaaS Platform <i>End-of-Module Project</i>	Feb 2025 — Jun 2025
- Developed a bilingual recruitment web application using microservices architecture. - Integrated open-source AI tools for CV parsing and automated professional response generation. - Containerized services with Docker for modular deployment and scalability.	

EDUCATION

Master of Data Science & Big Data, <i>Faculté des Sciences Ben M’sik - UH2C - Casablanca</i>	2024 - Present
Bachelor of Computer Science and Mathematics, <i>Faculté des Sciences Ben M’sik - UH2C - Casablanca</i>	2020 - 2023

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Nizar Belaatik

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CERTIFICATIONS

Data Science ALX Africa	<i>Jan 2024 - Present</i>
Generative AI for Everyone DeepLearning.AI	<i>Aug 2025</i>
Data Analytics ALX Africa — Credential ID: JM2TYSR86E	<i>Aug 2025</i>
Neural Networks and Deep Learning DeepLearning.AI	<i>Jul 2025</i>
Machine Learning with Python IBM	<i>Jun 2025</i>
Introduction to Data Engineering IBM	<i>May 2025</i>
Python for Data Science, AI & Development IBM	<i>May 2025</i>
Professional Foundations ALX Africa	<i>Apr 2025</i>
Scrum Foundation (SFPC™) Certiprof	<i>May 2024</i>

ACTIVITIES

Participated in organizing and monitoring university entrance exams and recruitment competitions in collaboration with Brain Management.